Zoe Lee

mechanical engineer · roboticist · aerospace engineer · researcher
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> PARSEC SURF Fellow | FSAE Chassis Lead | GA JSHS Finalist

EDUCATION

2025 – present — **California Institute of Technology (Caltech)**
B.S. Mechanical Engineering

2023 – 2025 — **Georgia Institute of Technology**
High School Dual Enrollment Program, Distance Math

2021 – 2025 — **Gwinnett School of Mathematics, Science, and Technology**
High School Diploma

courses — Experimental Robotics · Mechatronics & Electronic Systems · Mechanical Prototyping · Linear Algebra · Differential Equations · Software Design in C

ENGINEERING EXPERIENCE

Oct 2025 – present — **Thrust Vector Control Responsible Engineer | Dr. Judith Goodstein SURF Fellow**
Propulsion, Aeronautics, & Rocket Systems Engineering at Caltech
// mechatronics · inverse kinematics · actuation · systems integration
› Own the mechatronic TVC system for a self-landing rocket vehicle.
› Designed actuator mounts for rapid engine gimballing via inverse kinematics.
› Integrated mechanical, electrical, and control design through build and test.

Aug 2026 – present — **Chassis Subteam Lead**
Caltech Racing, Formula SAE
// solidworks · fea · structural design · fabrication
› Lead design and fabrication of the space-frame chassis all subsystems mount to.
› Authored and taught SOLIDWORKS + FEA onboarding training program.

Sep 2026 – present — **Mechanical Prototyping Teaching Assistant**
Caltech Jim Hall Design & Prototyping Laboratory
// machining · rapid prototyping · mentorship
› Coach students through CAD, machining, and laser and waterjet fabrication.

RESEARCH & ROBOTICS PROJECTS

2023 – 2025 — **Flash Joule Heating for Plastic Upcycling**
Independent Research | AP Research | GA JSHS Finalist
// pulsed power · circuit design · plasma · materials synthesis
› Built a pulsed-power rig from supercapacitor banks and custom circuits.
› Generated plasma to flash-convert plastic waste into graphene for clean energy.
› Led it solo: research question → apparatus → experiments → state symposia.

2023 – 2025 — **Team Leader; FRC Project Manager & Design Lead**
VEX 675 Zenith / FIRST Robotics Competition
// inventor · rapid prototyping · mechanism design · leadership
› Led mechanical design and fabrication of robots; 2× World Championship qualifier.
› Iterated robotic mechanisms through CAD, 3D printing, and build-test loops.
› Founded the school's first girls-focused robotics team.

Summer 2024 — **Autonomous Personal-Assistance Robot**
Georgia Governor's Honors Program | Engineering Major Finalist
// onshape · python · opencv · computer vision · autonomy
› Built an autonomous robot with a custom omnidirectional chassis and arms.
› Wrote OpenCV object recognition in Python, robust to object orientation.

SKILLS

robotics — Inverse kinematics, TVC & actuation, ROS, OpenCV, Altium PCB, soldering, UAVs
design — SOLIDWORKS, Onshape, Inventor, FEA, 3D printing, mill, lathe, waterjet, laser
software — Python, C, C++, MATLAB, Java, Dart, LaTeX, Linux/Ubuntu